NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SC2006 – Software Engineering

Lab 4 Deliverables

Lab Group	SCE3
Team	SilverConnect
	Lucas Cheong Wai (U2421947B)
	Wu Songxin
Members	Nicodemus Noel Lee
	Sivakumar Magina
	Lim Jia Yi

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1. Black Box Testing

I. AuthController

Control class being tested - AuthController

The **AuthController** class handles all aspects of user authentication in the system, including both **user login** and **registration**

The AuthController serves as a central component that ensures secure and reliable **user authentication**. When a user logs in, the controller requests for their login credentials (email and password) and validates them to ensure that the input data is properly formatted. It then retrieves the corresponding hashed password stored in the database for that email and compares it with the user's input. If both values match, authentication is confirmed, and the user is redirected to the application's home screen. In this process, the AuthController manages essential credentials such as the email and password used during login.

Furthermore, the AuthController is also responsible for handling **user registration**. During registration, it collects and verifies all required details to ensure that the submitted data is accurate and complete. Before saving the user's password to the database, the controller uses a secure hashing algorithm to protect sensitive information.

Depending on the user's role, different information is required to sign up. For example, Volunteers must list their availability and Interests, whereas Caregivers must input in their relationship to and email of the elderly.

The data handled by the AuthController during the signup process typically includes **Role**, **Name, Email, Age, Phone Number, Preferred Language, Password** and **Confirm Password** along with role-specific details.

II. Equivalence Class and Boundary Value Testing

<u>Equivalence Class Testing</u>

To conduct our black-box testing we design test cases with input of members from different equivalence classes. Every member of an equivalence class is assumed to trigger the same behaviour is the component being tested. We have designed our test cases to cover each class, an equivalence class is defined when each input produces the same output.

Additionally we define valid and invalid equivalence classes:

A **valid equivalence class** has valid input and will produce a valid output when

tested

An **Invalid equivalence class** has invalid input and will trigger an error when tested

Boundary Value Testing

Boundary Value Testing makes use of equivalence classes. We test just below, at or above the valid and invalid boundaries of the input domain partitioned by our equivalence classes. As this is where most bugs occur, efficient testing can be done.

1. Login
   Valid Equivalence Class: Correct Usernames/Emails and Passwords
   Invalid Equivalence Class: Incorrect or missing Usernames/Emails and Passwords

2. Sign up - if role selected is Elderly

   Valid Equivalence Class: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm Password in correct formats

   Invalid Equivalence Class: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm Password in incorrect formats or missing

3. Sign up - if role selected is Caregiver

   Valid Equivalence Class: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm Password, Elderly registered Email, Relation to Elderly in correct formats

   Invalid Equivalence Class: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm Password, Elderly registered Email, Relation to Elderly in incorrect formats or missing

4. Sign up - if role selected is volunteer

   Valid Equivalence Class: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm

Password, Preferred Activities, Availability, Travel Distance (optional) in correct formats

Invalid Equivalence Class: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm Password, Preferred Activities, Availability, Travel Distance (optional) in incorrect formats or missing

III. Test Cases and Results

a) Login

Input parameters: Email/Username and Password

No.	Test Input	Expected Output	Actual Output	Pass?
1	**(Valid)** Name: “Name” **(Valid)** Password: “password”	“Login Successful” Redirected to home screen	“Login Successful” Redirected to home screen	Yes
2	**(Valid)** Email: “e@gmail.com” **(Valid)** Password: “password”	“Login Successful” Redirected to home screen	“Login Successful” Redirected to home screen	Yes
3	**(Invalid)** Email: “ee@gmail.com” **(Valid)** Password: “password”	Login Failed, Invalid username /email or password	Login Failed, Invalid username /email or password	Yes
4	**(Valid)** Email: “ee@gmail.com” **(Invalid)** Password: “”	Login Failed, Invalid username /email or password “Please fill in this field” notification appears beneath Password field	Login Failed, Invalid username /email or password “Please fill in this field” notification appears beneath Password field	Yes

5	**(Invalid)** Email: “” **(Valid)** Password: “password”	Login Failed, Invalid username /email or password “Please fill in this field” notification appears beneath Email field	Login Failed, Invalid username /email or password “Please fill in this field” notification appears beneath Email field	Yes
6	**(Invalid)** Email: “” **(Invalid)** Password: “”	Login Failed, Invalid username /email or password “Please fill in this field” notification appears beneath Email field	Login Failed, Invalid username /email or password “Please fill in this field” notification appears beneath Email field	Yes

b) Sign up - if role selected is Elderly

Input parameters: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm Password

No.	Test Input	Expected Output	Actual Output	Pass?
1	**(All valid inputs)** Role: Elderly Name: “Name” Email: “e@gmail.com” Age: “72” Phone number: “42149835” Preferred Language: “English” Password: “password” Confirm Password: “password”	Successful Signup	Successful Signup	Yes
2	**(All valid inputs**	“Please fill in this	“Please fill in this	Yes

	except Name) Name: “”	field” notification appears beneath Name	field” notification appears beneath Name	
3	**(All valid inputs except Email)** Email: “”	“Please fill in this field” notification appears beneath Email	“Please fill in this field” notification appears beneath Email	Yes
4	**(All valid inputs except Email)** Email: “e”	“Please include an ‘@’ in the email address.”	“Please include an ‘@’ in the email address.”	Yes
5	**(All valid inputs except Email)** Email: “e@”	“Please enter a part after ‘@’.”	“Please enter a part after ‘@’.”	Yes
6	**(All valid inputs except Email)** Email: “e@pea”	“Invalid email format”	“Invalid Email format”	Yes
7	**(All valid inputs except Age)** Age: “”	“All users must provide age, phone number and preferred language” notification appears	“All users must provide age, phone number and preferred language” notification appears	Yes
8	**(All valid inputs except Phone Number)** Phone Number: “”	“All users must provide age, phone number and preferred language” notification appears	“All users must provide age, phone number and preferred language” notification appears	Yes
9	**(All valid inputs except Preferred Language)** Preferred Language: “”	“All users must provide age, phone number and preferred language” notification appears	“All users must provide age, phone number and preferred language” notification appears	Yes
10	**(All valid inputs except Password)** Password: “”	“Please fill in this field” notification appears beneath Password field	“Please fill in this field” notification appears beneath Password field	Yes
11	**(All valid inputs except Password)** Password: “Pass”	“Password must be at least 6 characters”	“Password must be at least 6 characters”	Yes
12	**(All valid inputs except Confirm Password)** Confirm Password: “”	“Please fill in this field” notification appears beneath Confirm Password field	“Please fill in this field” notification appears beneath Confirm Password field	Yes

13	**(All valid inputs except Password and Confirm Password)** Password: “A” Confirm Password: “B”	Passwords do not match	Passwords do not match	Yes

c) Sign up - if role selected is Caregiver

Input parameters: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm Password, Elderly registered Email, Relation to Elderly

No.	Test Input	Expected Output	Actual Output	Pass?
1	**(All valid inputs)** Role: Caregiver Name: Name Email: f@gmail.com Age: 34 Phone number: 12345678 Language: English Password: Password Confirm Password: Password Elderly Registered Email: e@gmail.com Relation to elderly: son	Successful Signup	Successful Signup	Yes
2	**(All valid inputs except Elderly Registered Email)** Elderly Registered Email: “”	Caregiver must provide Elderly email and relationship	Caregiver must provide Elderly email and relationship	Yes
3	**(All valid inputs except Relationship to Elderly)** Relationship to Elderly: “”	Caregiver must provide Elderly email and relationship	Caregiver must provide Elderly email and relationship	Yes

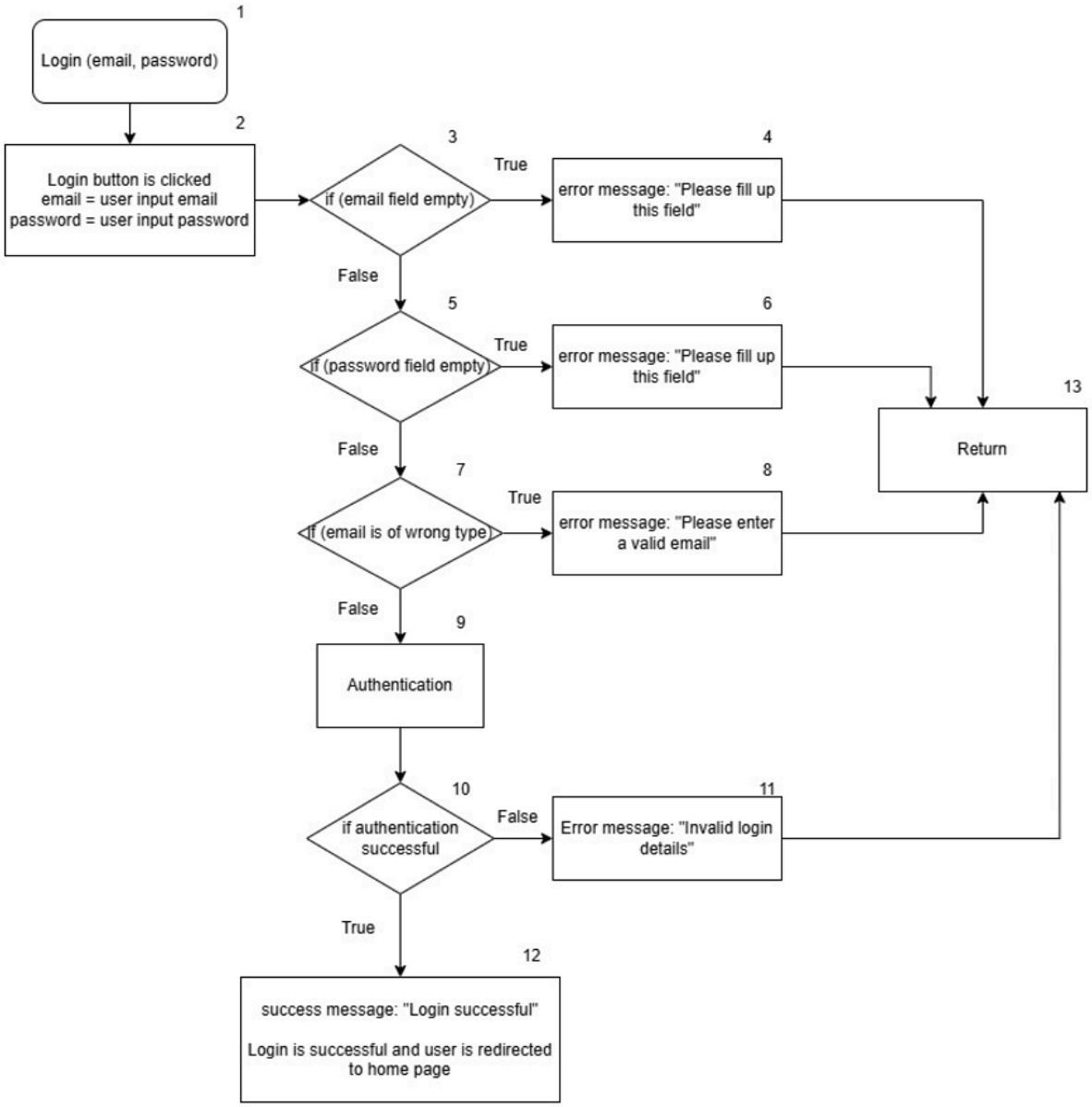
d) Sign up - if role selected is Volunteer

Input parameters: Role, Name, Email, Age, Phone Number, Preferred Language, Password, Confirm Password, Preferred Activities, Availability, Travel Distance (optional)

No.	Test Input	Expected Output	Actual Output	Pass?
1	**(All valid inputs)** Role: Volunteer Name: Name Email: v@gmail.com Age: 27 Phone number: 12345678 Language: English Password: Password Confirm Password: Password Preferred Activities: Mahjong Availability: Mon Am Travel Distance (km): 5	Successful Signup	Successful Signup	Yes
2	**(All valid inputs except Preferred Activities)** Preferred Activities: “”	Volunteer must provide activities and availability	Volunteer must provide activities and availability	Yes
3	**(All valid inputs except Availability)** Availability: “”	Volunteer must provide activities and availability	Volunteer must provide activities and availability	Yes
4	**(All valid inputs except Travel Distance)** Travel Distance: “”	Successful Signup	Successful Signup	Yes

2. White Box Testing (2 methods only)

I. Login



I.I Control Flow Graph

<u>Cyclomatic Complexity</u> = Decision points + 1 = 4 + 1 = 5

I.II Basic Path Testing

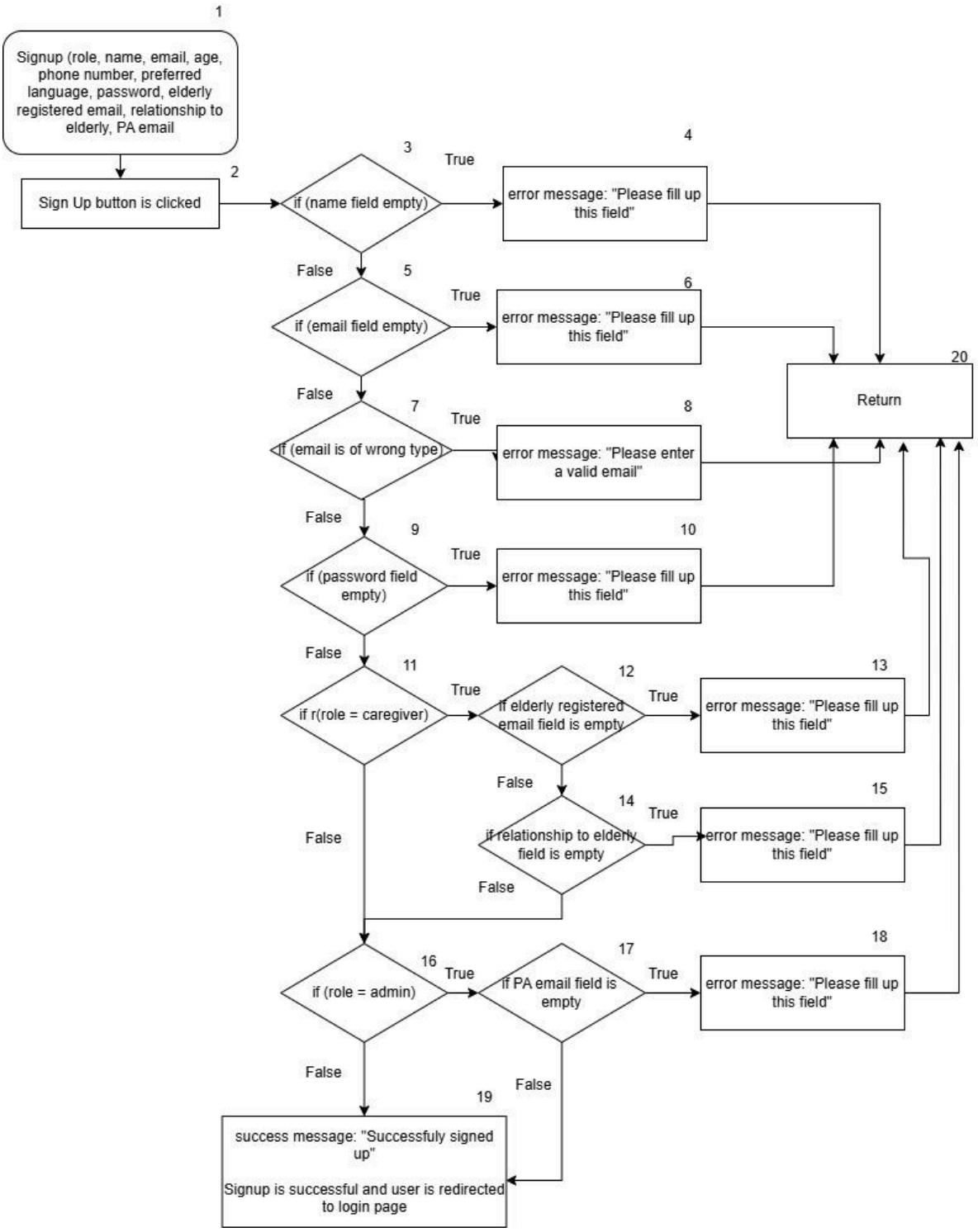
Basis paths

1. Baseline path: 1,2,3,5,7,9,10,12
2. Basis path 2: 1,2,3,4,13
3. Basis path 3: 1,2,3,5,6,13
4. Basis path 4: 1,2,3,5,7,8,13
5. Basis path 5: 1,2,3,5,7,9,10,11,13

I.III Test Cases and Results

No.	Test Input	Expected Output	Actual Output	Pass?
1	Email =”f@gmail.com” Password = “password”	Successful login	Successful login	Yes
2	Email = “” Password = “password”	“Please fill up this field”	“Please fill up this field”	Yes
3	Email= “f@gmail.com” Password = “”	“Please fill up this field”	“Please fill up this field”	Yes
4	Email = “a” Password = “password”	“Please enter a valid email address”	“Please enter a valid email address”	Yes
5	Email = “f@gmail.com” Password = “1234”	“Invalid login details”	“Invalid login details”	Yes

II. Signup



II.I Control Flow Graph

<u>Cyclomatic Complexity</u> = Decision points + 1 = 9 + 1 = 10

II.II Basic Path Testing

<u>Basis paths</u>

1. Baseline path: 1,2,3,5,7,9,11,16,19
2. Basis path 2: 1,2,3,5,7,9,11,12,14,16,19
3. Basis path 3: 1,2,3,5,7,9,11,16,17,19
4. Basis path 4: 1,2,3,4,20
5. Basis path 5: 1,2,3,5,6,20
6. Basis path 6: 1,2,3,5,7,8,20
7. Basis path 7: 1,2,3,5,7,9,10,20
8. Basis path 8: 1,2,3,5,7,9,11,12,13,20
9. Basis path 9: 1,2,3,5,7,9,11,12,14,15,20
10. Basis path 10: 1,2,3,5,7,9,11,16,17,18,20

II.III Test Cases and Results

No.	Test Input	Expected Output	Actual Output	Pass?
1	Role: Elderly Name: Name Email: e@gmail.com Age: 72 Phone number: 42149835 Language: English Password: password Confirm Password: password	Successfully signed up	Successfully signed up	Yes
2	Role: Caregiver Name: Name Email: f@gmail.com Age: 34 Phone number: 12345678 Language: English Password: password Confirm Password: Password Elderly registered email: e@gmail.com Relation to elderly: son	Successfully signed up	Successfully signed up	Yes
3	Role: Admin Name: Name Email: h@gmail.com Age: 37	Successfully signed up	Successfully signed up	Yes

	Phone number: 12345678 Language: English Password: password Confirm Password: Password PA email: h@pa.gov.sg			
4	Role: Elderly Name: Email: e@gmail.com Age: 72 Phone number: 42149835 Language: English Password: password Confirm Password: password	“Please fill up this field”	“Please fill up this field”	Yes
5	Role: Elderly Name: Name Email: Age: 72 Phone number: 42149835 Language: English Password: password Confirm Password: password	“Please fill up this field”	“Please fill up this field”	Yes
6	Role: Elderly Name: Name Email: e Age: 72 Phone number: 42149835 Language: English Password: password Confirm Password: password	“Please enter a valid email address”	“Please enter a valid email address”	Yes
7	Role: Elderly Name: Name Email: e@gmail.com Age: 72 Phone number: 42149835 Language: English Password:	“Please fill up this field”	“Please fill up this field”	Yes

	Confirm Password: password			
8	Role: Caregiver Name: Name Email: f@gmail.com Age: 34 Phone number: 12345678 Language: English Password: password Confirm Password: Password Elderly registered email: Relation to elderly: son	“Please fill up this field”	“Please fill up this field”	Yes
9	Role: Caregiver Name: Name Email: f@gmail.com Age: 34 Phone number: 12345678 Language: English Password: password Confirm Password: Password Elderly registered email: e@gmail.com Relation to elderly:	“Please fill up this field”	“Please fill up this field”	Yes
10	Role: Admin Name: Name Email: h@gmail.com Age: 37 Phone number: 12345678 Language: English Password: password Confirm Password: Password PA email:	“Please fill up this field”	“Please fill up this field”	Yes